MSc Data Science
DS601: Mathematical Foundations for Data Science
Date: 24/08/2026

Tutorial 5

Question 1 — Random Variable

A fair coin is tossed three times. Let Xdenote the number of heads obtained.

(a) Define the random variable $X$and list all possible values of X.
(b) Construct the probability distribution of X.
(c) Find $P(X \geq 2)$.
(d) Find $E[X]$.

Question 2 — Probability Distribution

A discrete random variable Xhas probability mass functions

$$P(X = x) = k(x + 1),\ x = 0, 1, 2, 3, 4.$$

Find the value of $k$and hence determine the complete probability distribution.

Also find:

(a) $P(1 < X \leq 3)$
(b) $P(X \geq 2)$
(c) $E[X]$
(d) $Var(X)$

Question 3 — PMF, PDF and CDF

(A) Continuous Random Variable

Let Xbe uniformly distributed over (10).

(a) Find the PDF of X.
(b) Find the CDF of X.
(c) Calculate $P(2 < X < 7)$.
(d) Verify the result using the CDF.

Question 4 — Binomial Distribution

A basketball player takes 12 free-throw shots in a match. The probability of successfully making each shot is 0.75, and the shots are independent. Let X be the number of successful shots.

(a) Identify the distribution of $X$and state its parameters.
(b) Find $P(X = 9)$.
(c) Find $P(X \geq 8)$.
(d) Given that the player makes at least 8 successful shots, find the probability that exactly 10 are successful.
(e) Find the smallest integer $r$such that $P(X \geq r) < 0.10$.

Question 5 — Uniform Distribution

The lifetime X, in hours, of a certain battery is uniformly distributed between 800 and 1200 hours.

(a) Find $P(X > 1050)$.
(b) Find $P(900 < X < 1100)$.
(c) Given that a battery has already lasted 900 hours, find the probability that it lasts at least another 200 hours.

(d) Find $a$such that $P(X < a) = 0.35$.

Question 6 — Function of a Random Variable

Let Xbe a continuous random variable with PDF

$$f_X(x) = \{2x,\ 0 < x < 1,\ 0,\ otherwise.$$

Define

$$Y = X^2.$$

(a) Find the CDF of Y.

(b) Derive the PDF of Y.

(c) Find $P\left(\frac{1}{4} < Y < \frac{3}{4}\right)$.

(d) Find $E[Y]$.